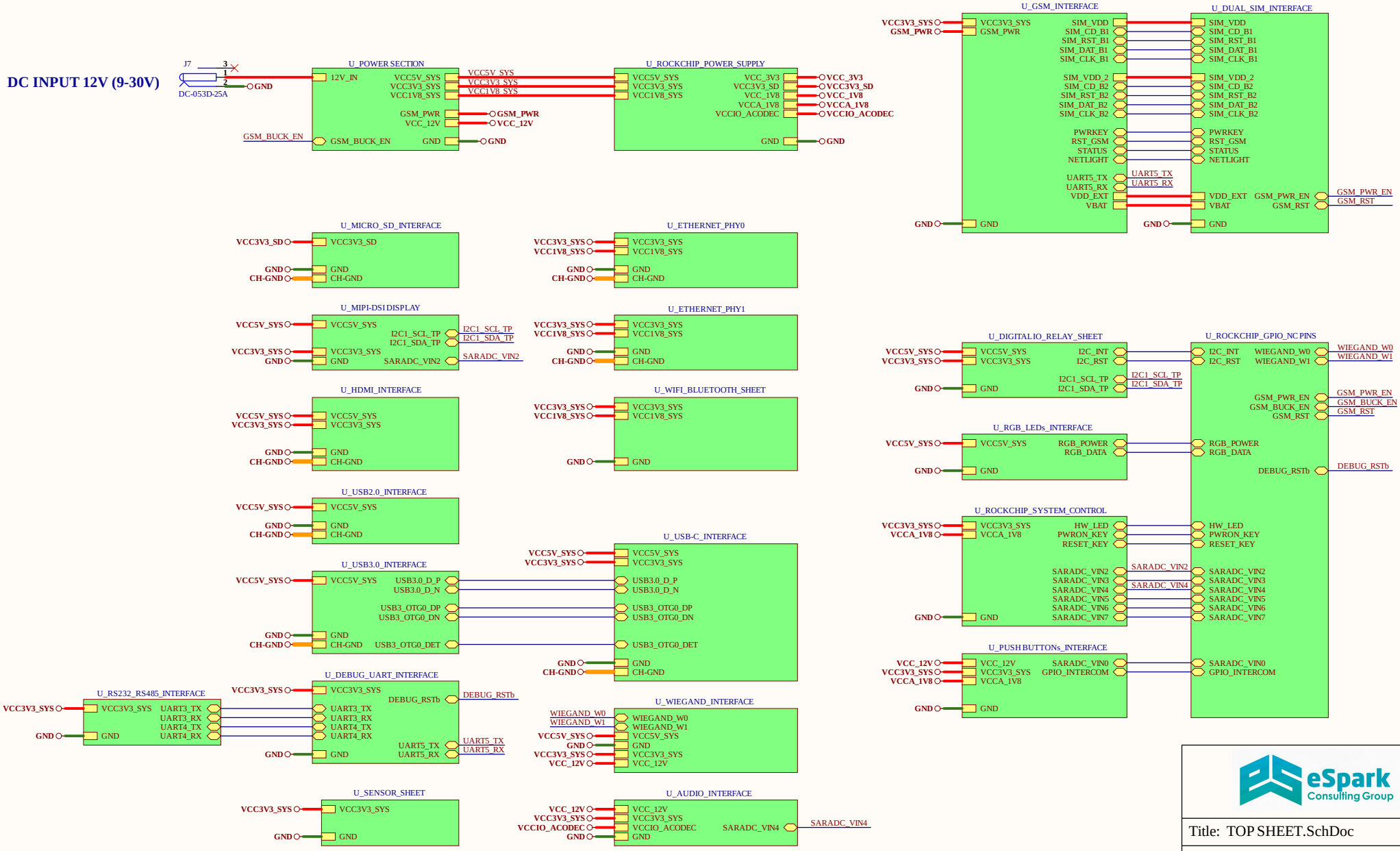
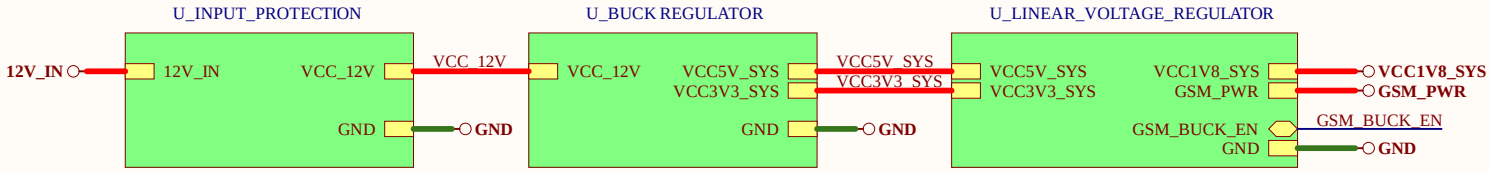
TOP SHEET

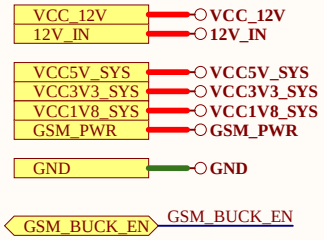


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Title: TOP SHEET.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 27

POWER SECTION

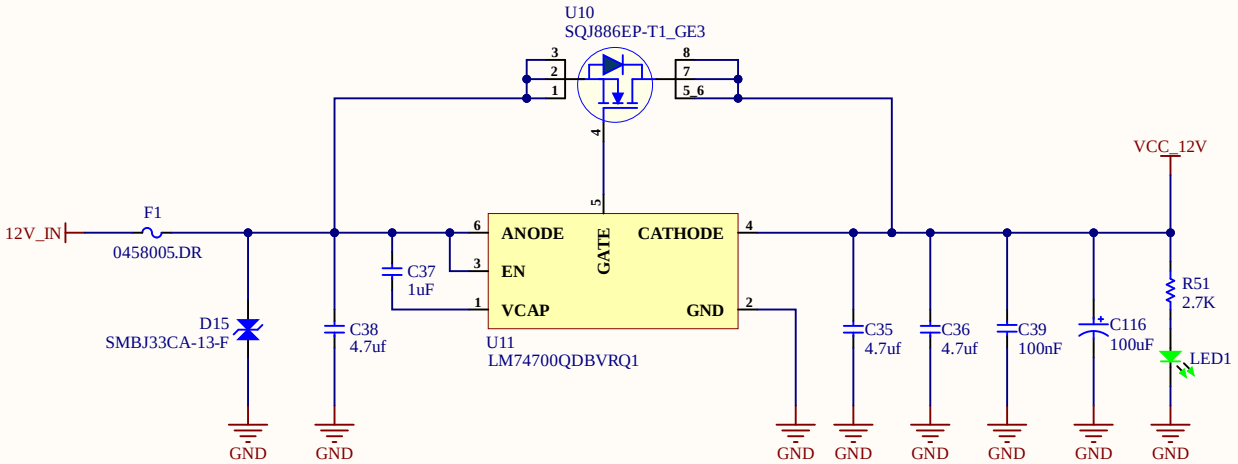




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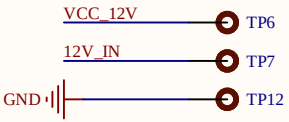
Title: POWER SECTION.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 27

INPUT PROTECTION CIRCUIT



REVERSE POLARITY PROTECTION

TEST POINTs

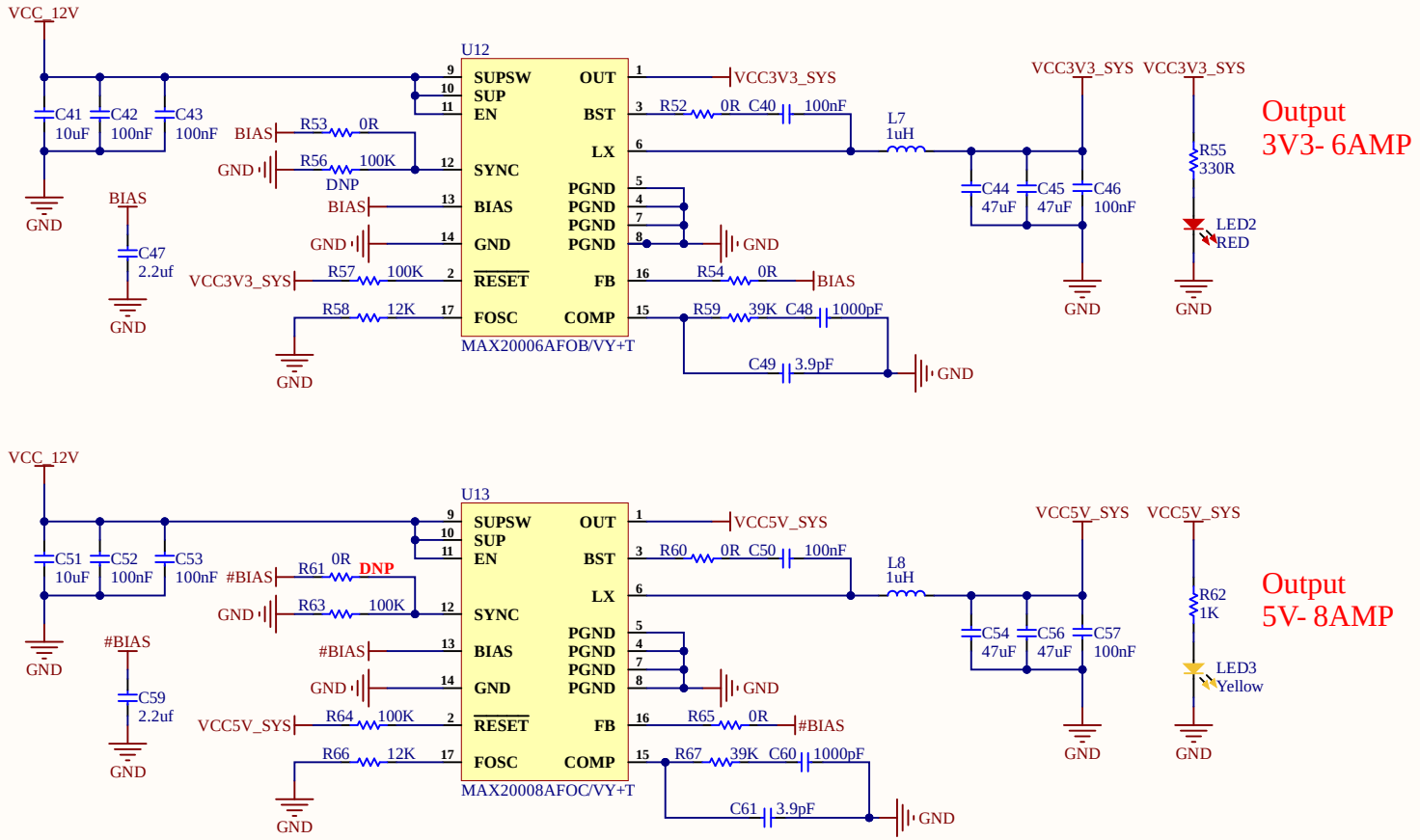




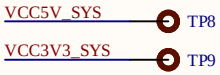
eSpark Consulting Group

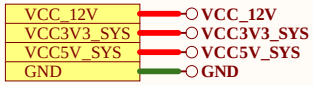
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File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 27

BUCK REGULATOR



TEST POINTs



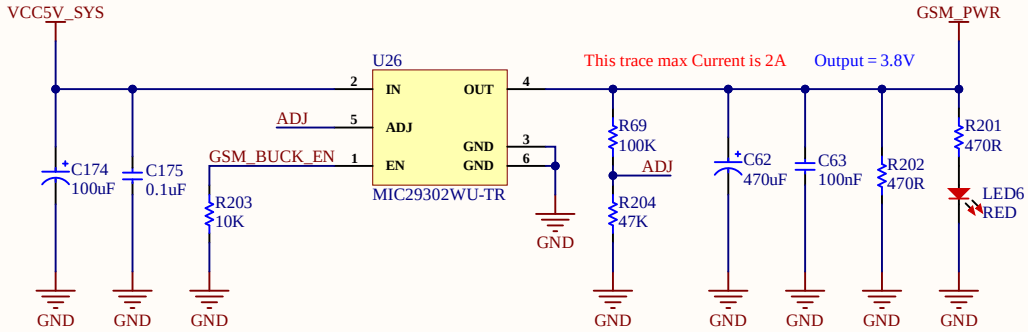


eSpark Consulting Group

Title: BUCK REGULATOR.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 27

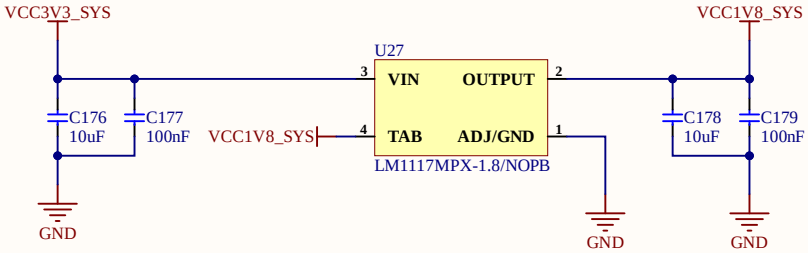
LINEAR_VOLTAGE_REGULATOR

LDO FOR GSM POWER



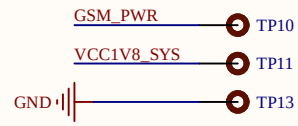
Output
3V8- 3AMP

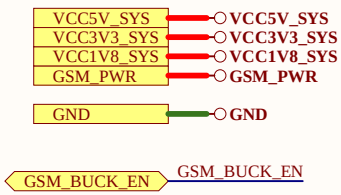
1V8 LDO



Output
1V8- 0.8AMP

TEST POINTs



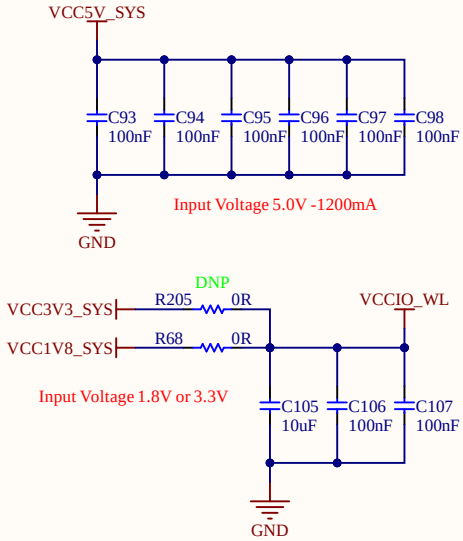


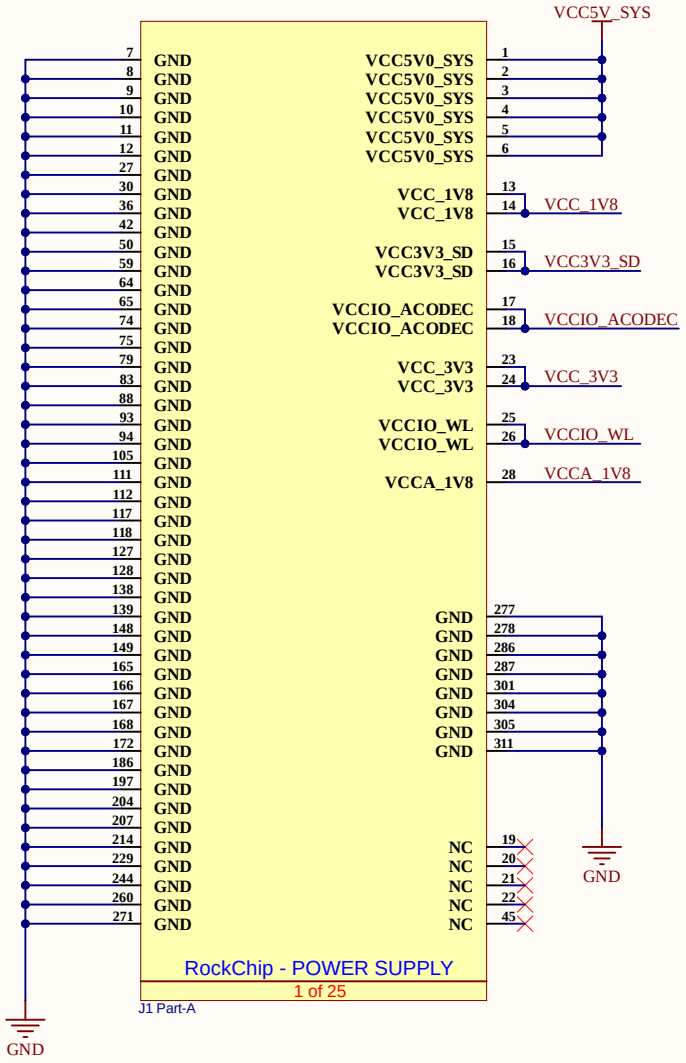
eSpark Consulting Group

Title: LINEAR_VOLTAGE_REG.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 24/05/2024	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 27

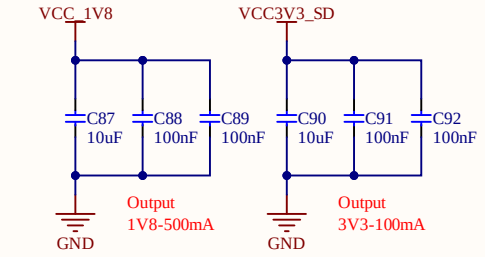
ROCKCHIP POWER SUPPLY

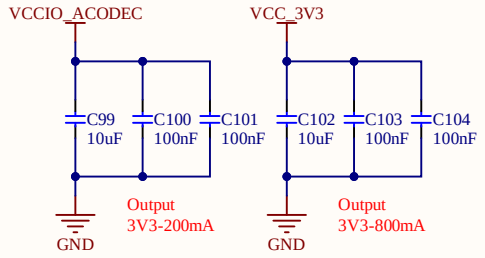
DECOUPLING CAPACITORs
PLACE CLOSE TO EACH PIN

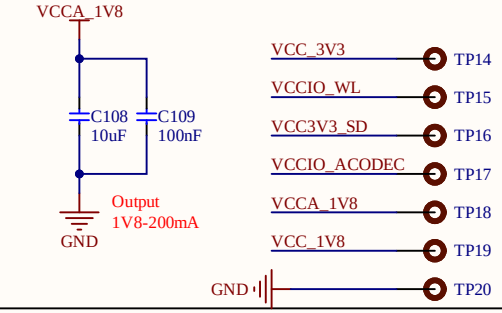




DECOUPLING CAPACITORs
PLACE CLOSE TO EACH PIN







VCC3V3_SYS — VCC3V3_SYS
VCC5V_SYS — VCC5V_SYS
GND — GND

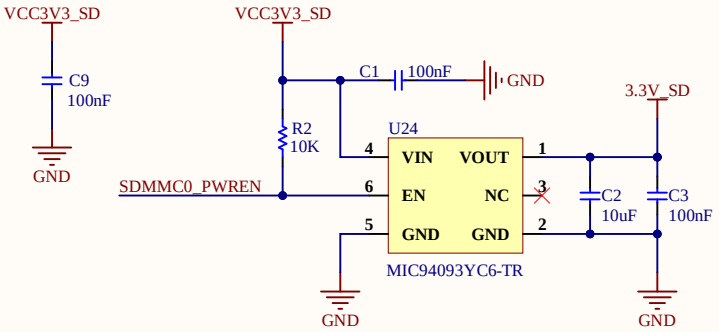
VCC_3V3 — VCC_3V3
VCC_1V8 — VCC_1V8
VCCA_1V8 — VCCA_1V8
VCC3V3_SD — VCC3V3_SD
VCC1V8_SYS — VCC1V8_SYS

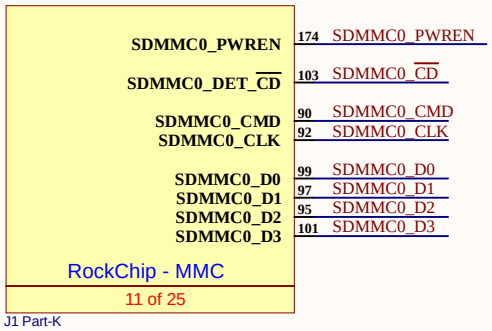
VCCIO_ACODEC — VCCIO_ACODEC

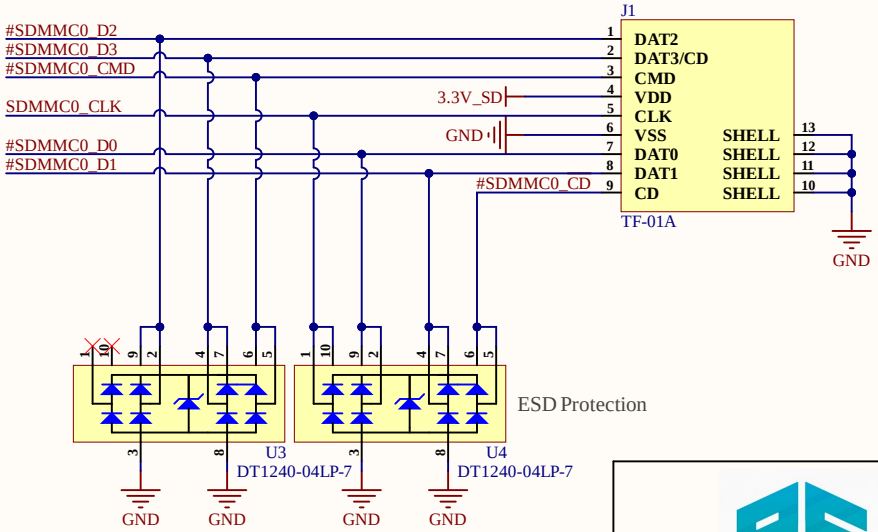
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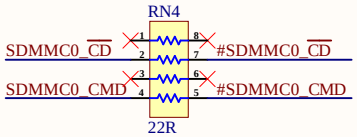
Title: ROCKCHIP_PWR_SUPPLY.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 27

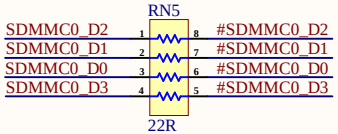
MICRO SD CARD INTERFACE

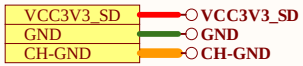








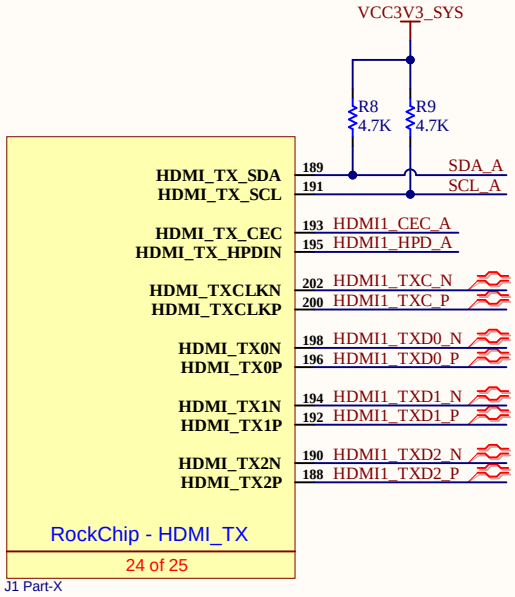


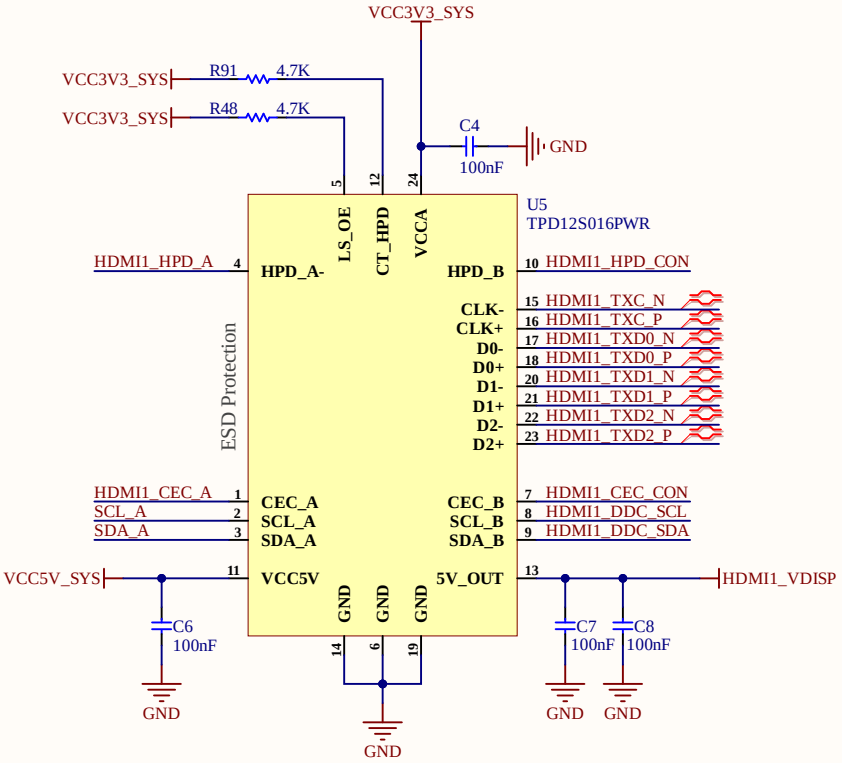


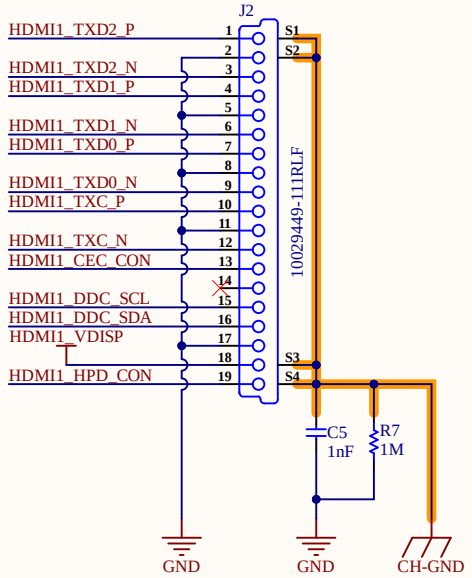
eSpark Consulting Group

Title: MICRO_SD_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page	Num: 7
Drawn By: Huzaifa Ali	Page	Total: 27

HDMI INTERFACE

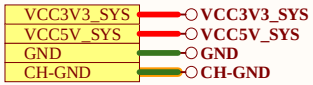






TEST POINTs

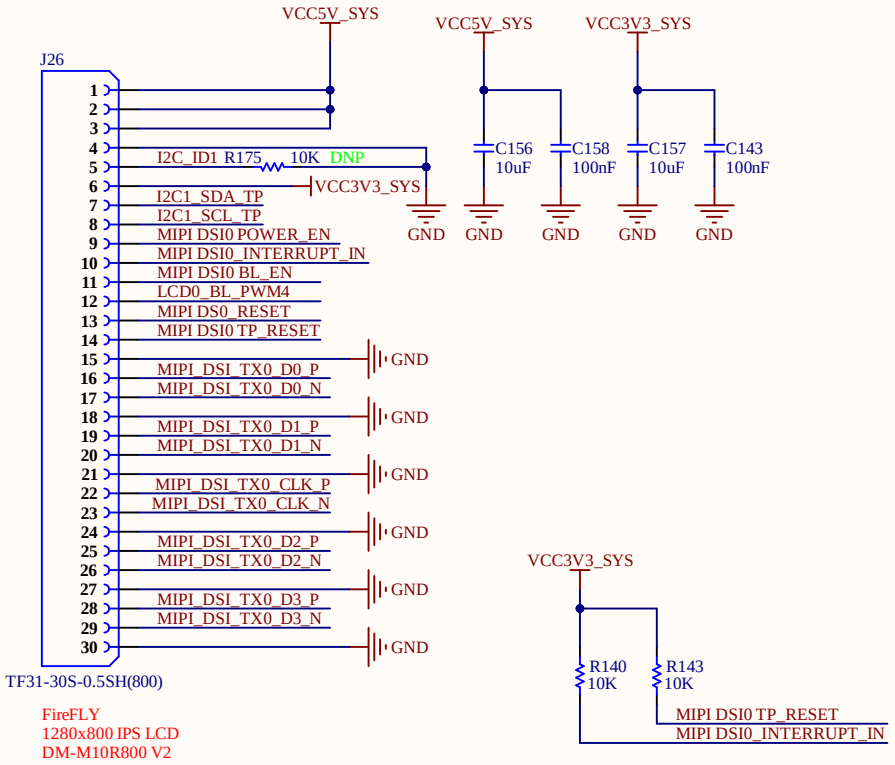


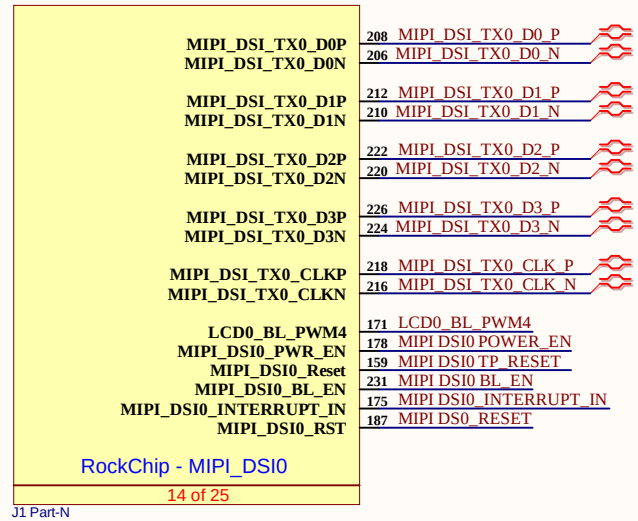


eSpark Consulting Group

Title: HDMI_INTERFACE.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 27

MIPI-DSI DISPLAY





SARADC_VIN2_LCD_ID_R116 0R I2C_ID1

I2C1_SDA_TP — I2C1_SDA_TP
I2C1_SCL_TP — I2C1_SCL_TP

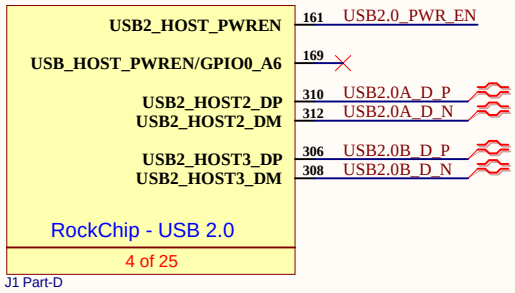
SARADC_VIN — SARADC_VIN2_LCD_ID

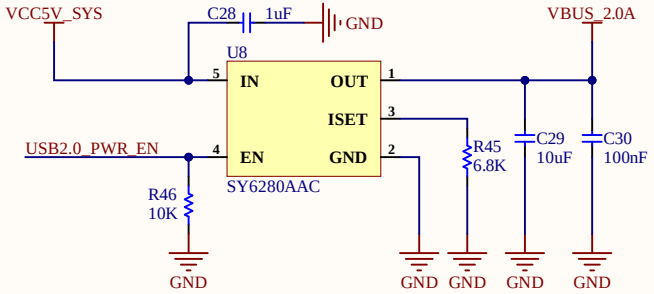
VCC3V3_SYS — VCC3V3_SYS
VCC5V_SYS — VCC5V_SYS
GND — GND

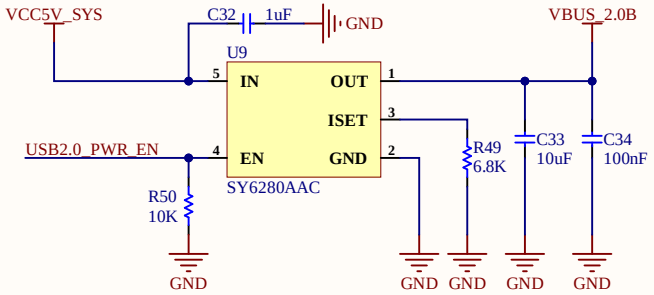
eSpark Consulting Group

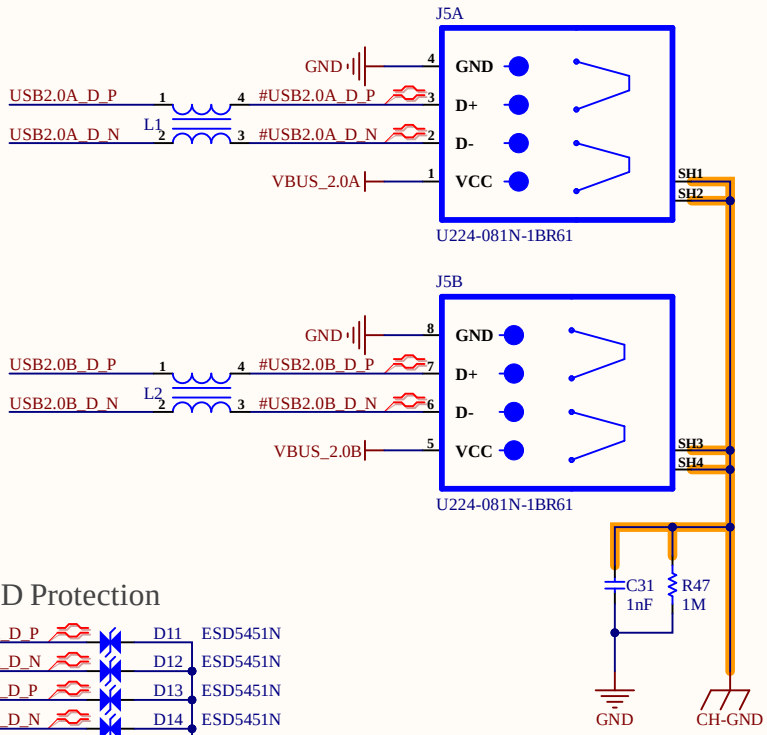
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File: Carrier-Board-v1.1.PrjPcb		
Date: 10/09/2024	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 27

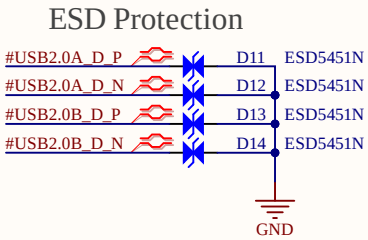
USB2.0 INTERFACE

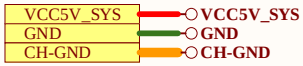








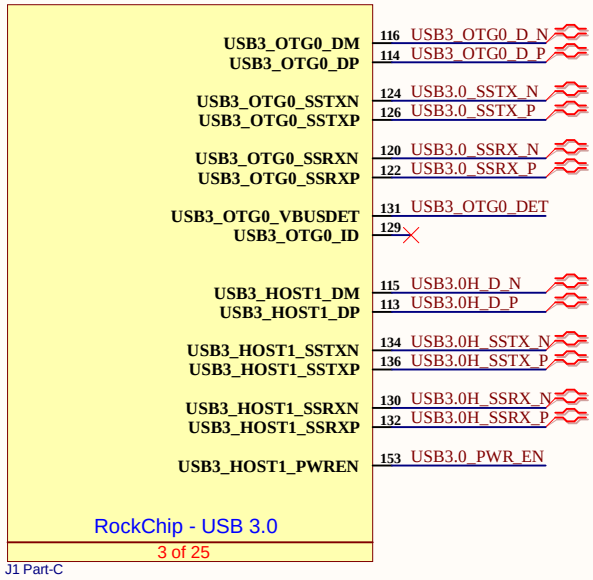


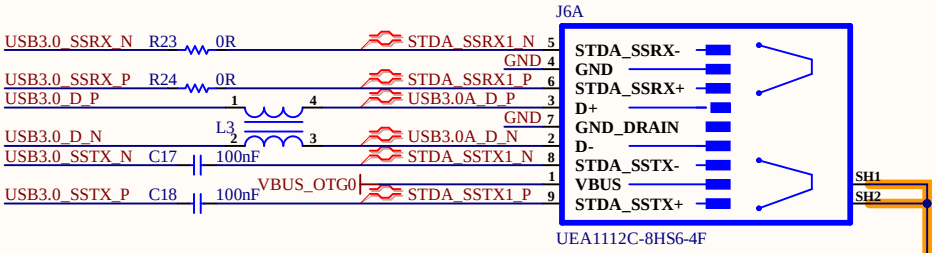


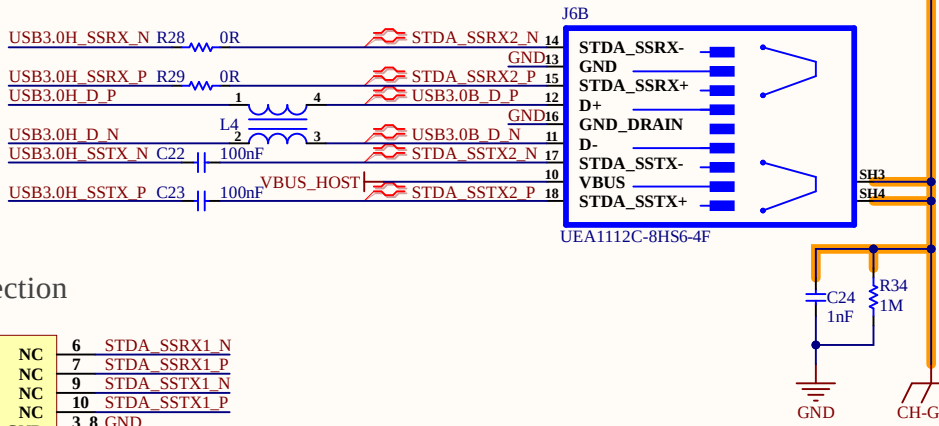
eSpark Consulting Group

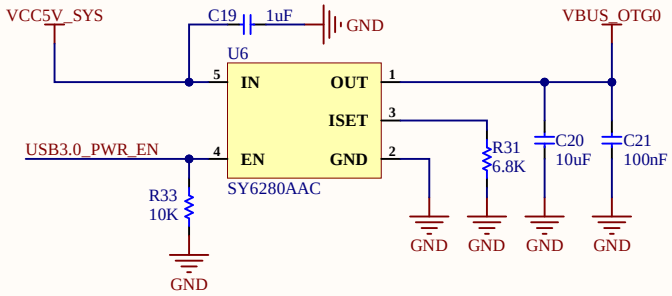
Title: USB2.0_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 20/05/2024	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 27

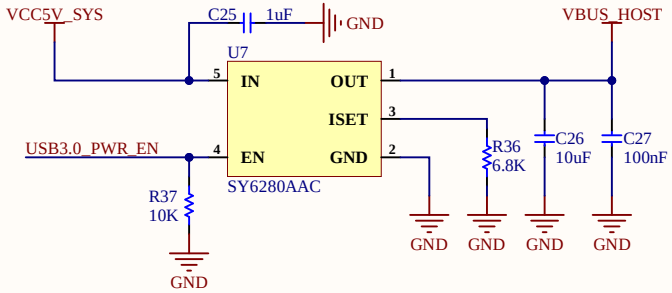
USB3.0 INTERFACE



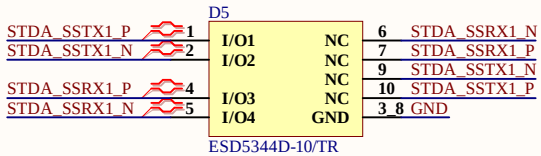


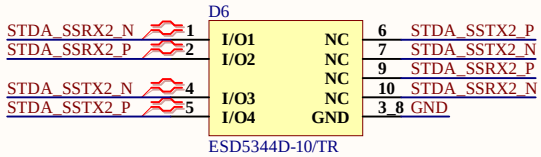


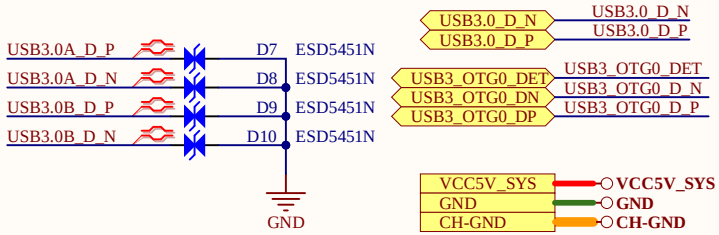




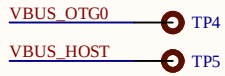
ESD Protection







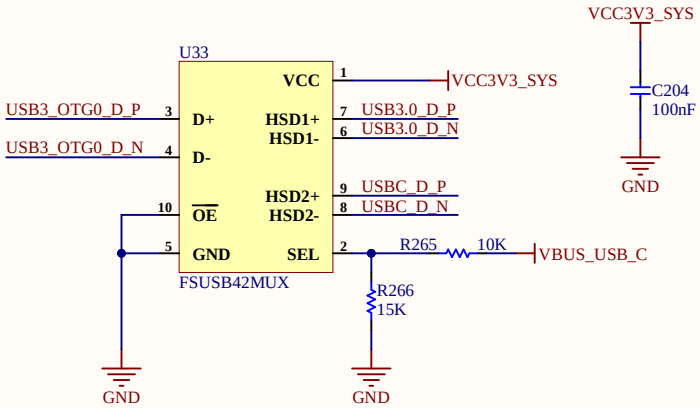
TEST POINTs

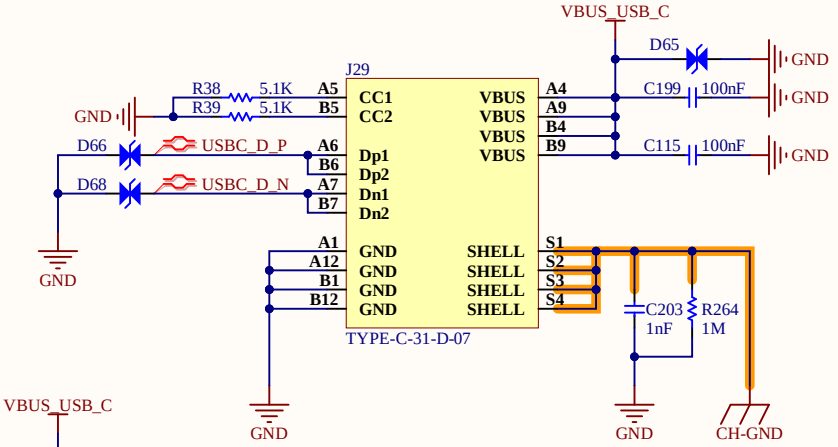


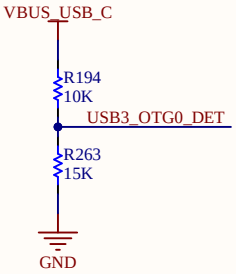
eSpark Consulting Group

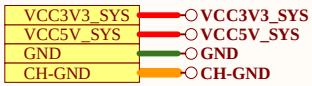
Title: USB3.0_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page Num:	11
Drawn By: Huzaifa Ali	Page Total:	27

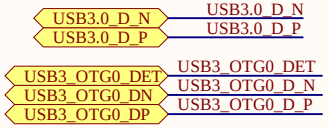
USB-C_INTERFACE







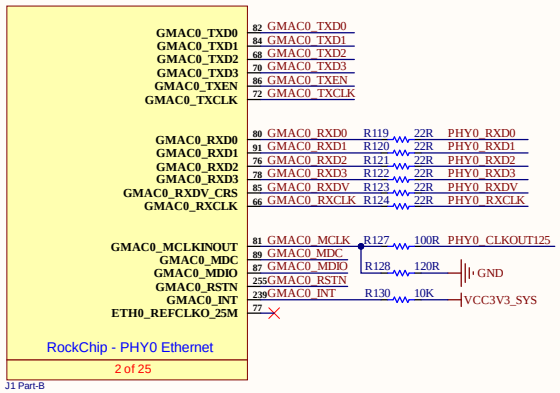


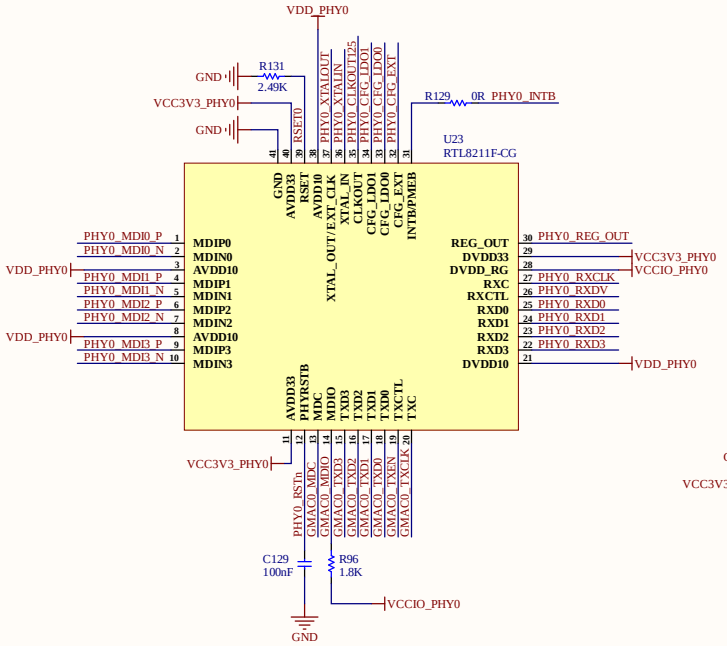


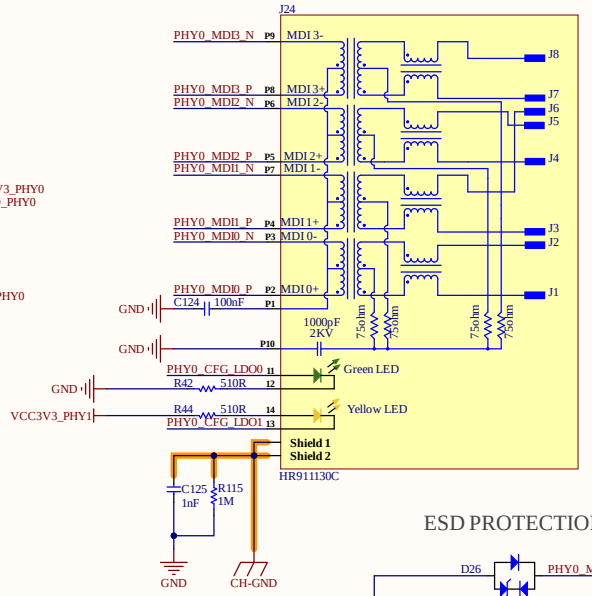
eSpark Consulting Group

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Date: 06/09/2024	Page Num:	12
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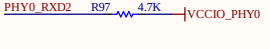
ETHERNET_PHY0_INTERFACE





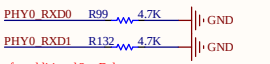


PLL MODE



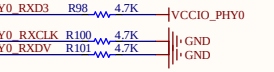
Pull-up to disable PLL

TXC/RXC DELAY CONFIG

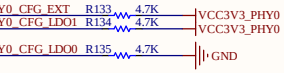


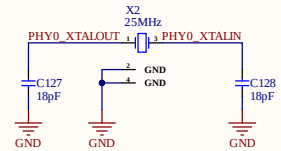
Pull-up for additional 2ns Delay

PHY0 ADDRESS CONFIG

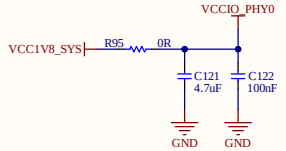


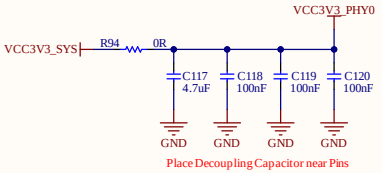
PHY0_IO Voltage Config

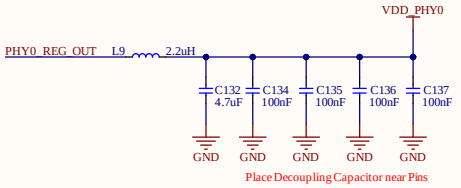




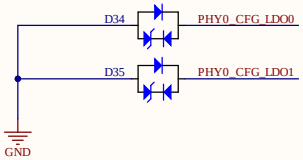


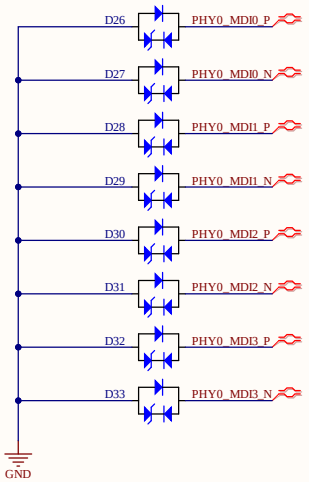


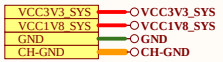




ESD PROTECTION

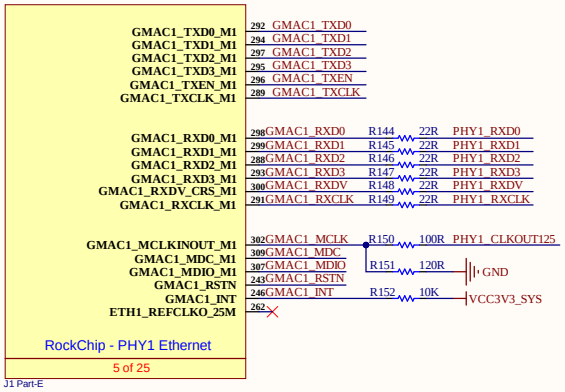


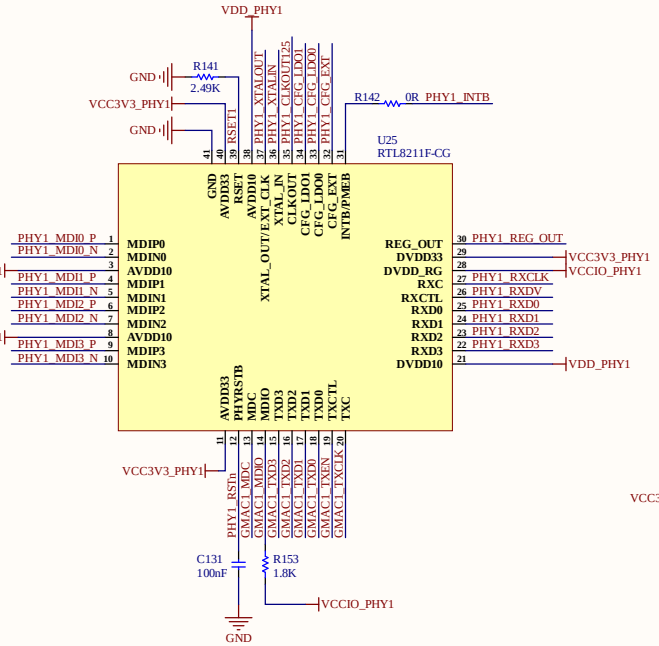


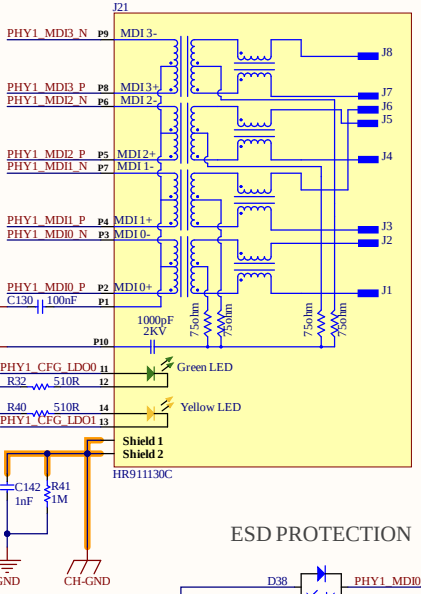


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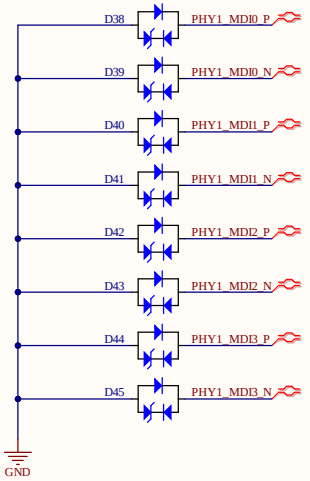
ETHERNET_PHY1_INTERFACE







ESD PROTECTION



PLL MODE

PHY1_RXD2 R156 4.7K VCCIO_PHY1

Pull-up to disable PLL

TXC/RXC DELAY CONFIG

PHY1_RXD0 R160 4.7K GND

PHY1_RXD1 R161 4.7K GND

Pull-up for additional 2ns Delay

PHY1 ADDRESS CONFIG

PHY1_RXD3 R165 4.7K VCCIO_PHY1

PHY1_RXCLK R166 4.7K GND

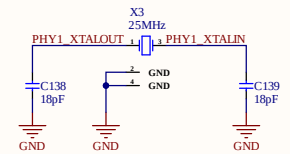
PHY1_RXDV R167 4.7K GND

PHY1_IO Voltage Config

PHY1_CFG_EXT R170 4.7K VCC3V3_PHY1

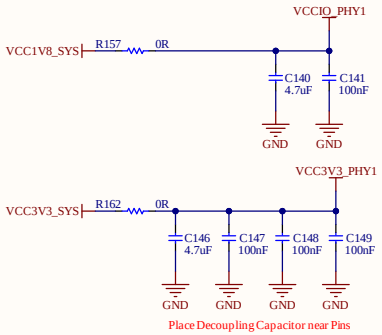
PHY1_CFG_LDO1 R171 4.7K VCC3V3_PHY1

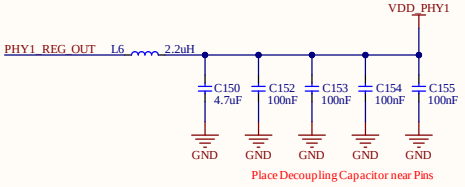
PHY1_CFG_LDO0 R172 4.7K GND

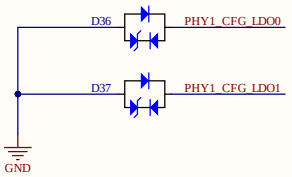


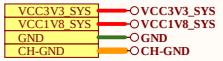
GMAC1_INT R126 0R PHY1_INTB

GMAC1_RSTN R136 0R PHY1_RSTn



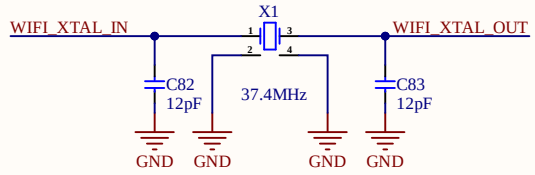


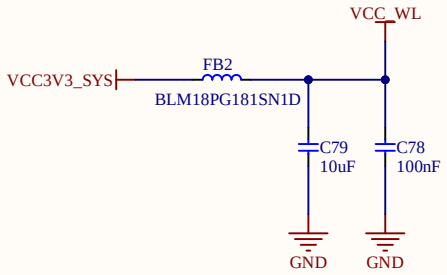




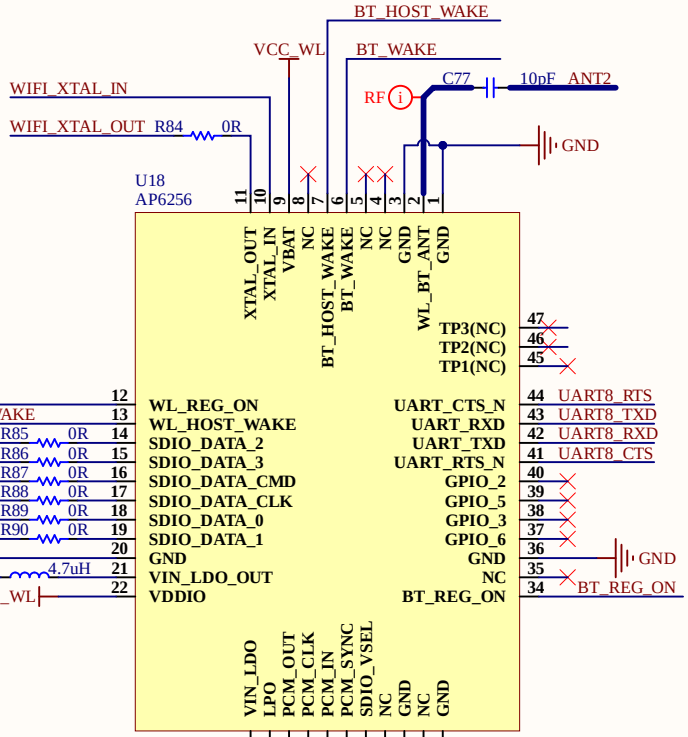
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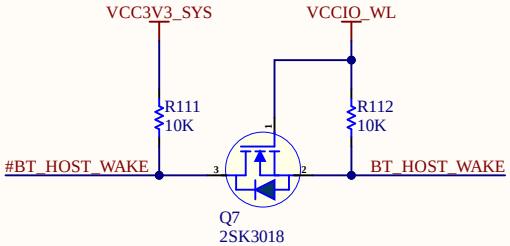
WIFI_BLUETOOTH_INTERFACE

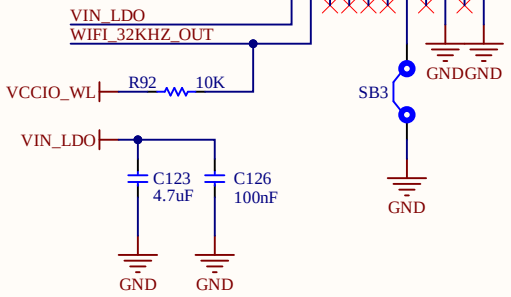


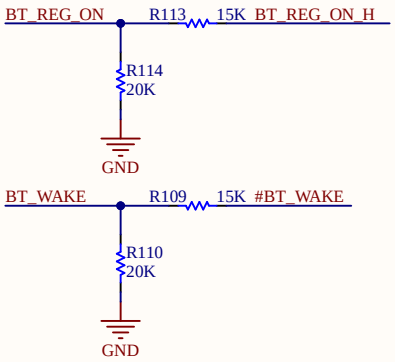


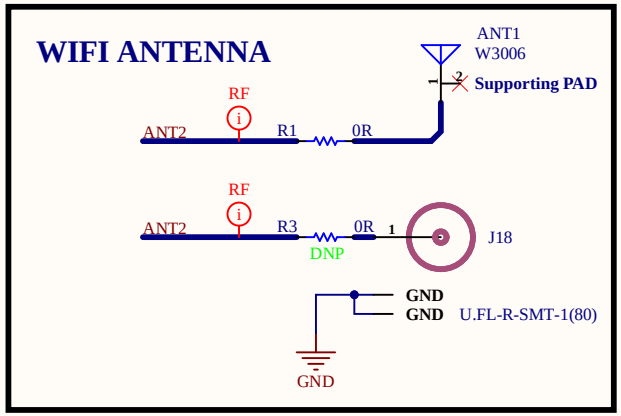
Dual Band WiFi

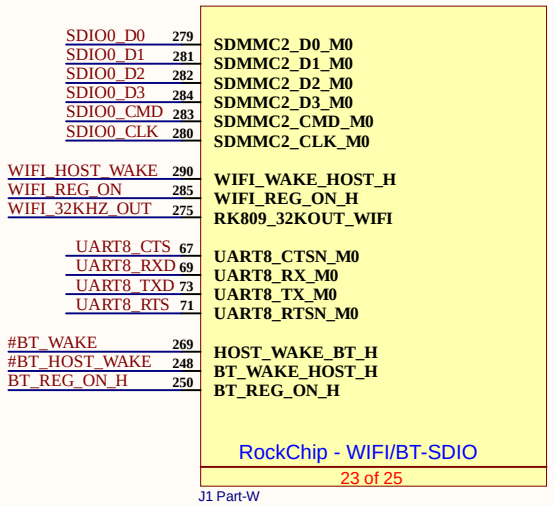


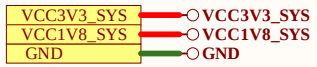






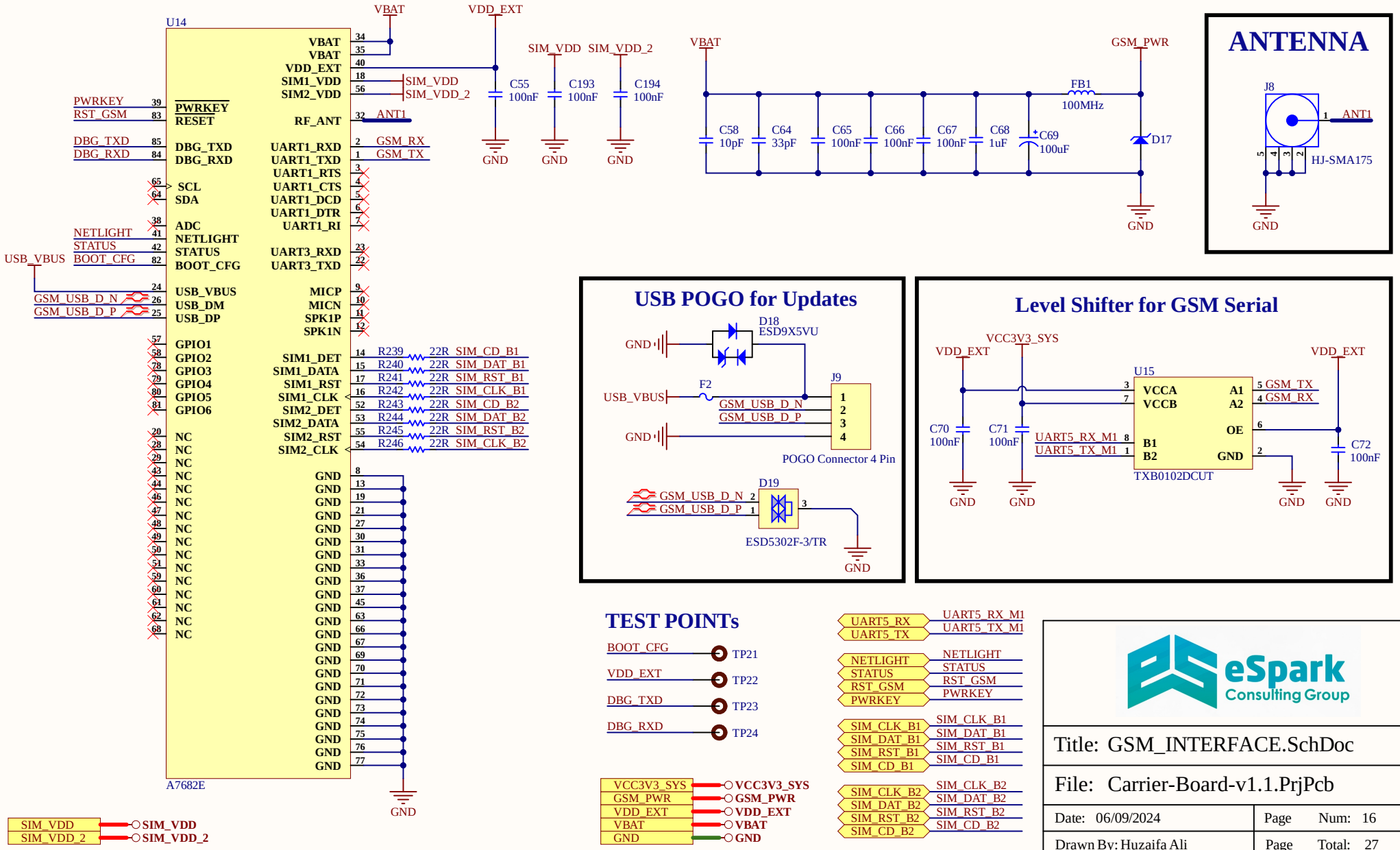






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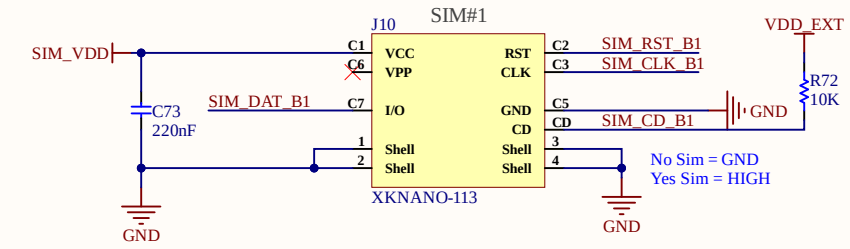
GSM_INTERFACE

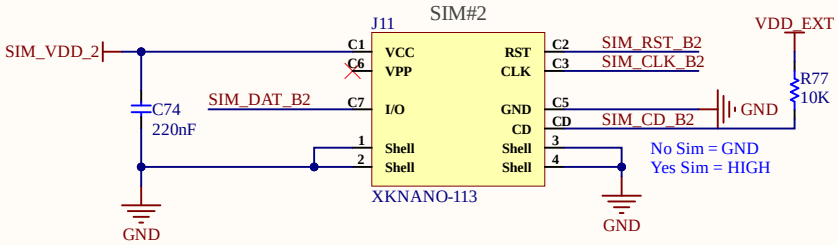


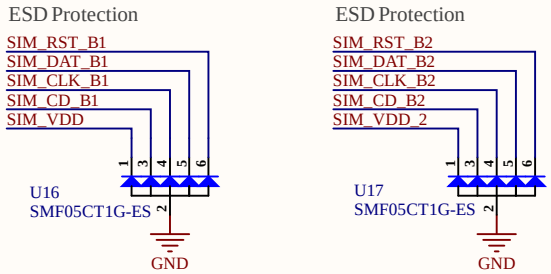
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DUAL_SIM_INTERFACE

DUALSIM WITH MULTIPLEXER

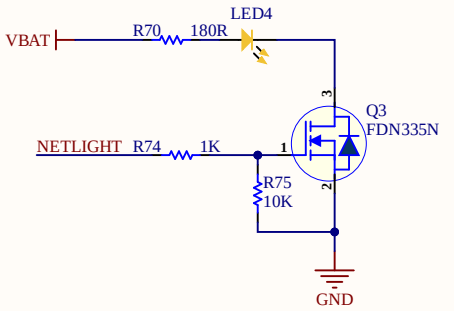


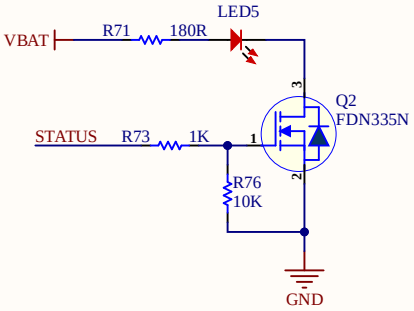


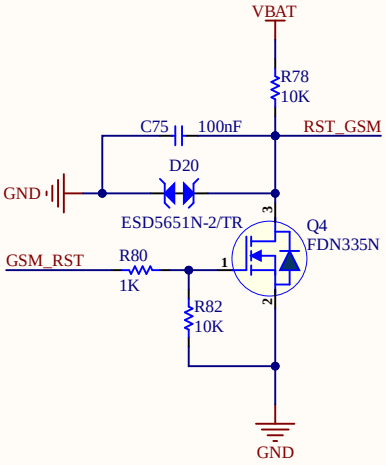


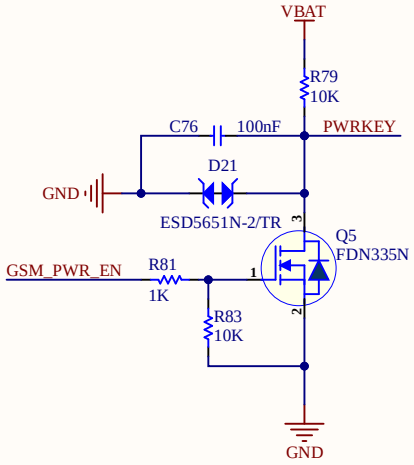
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

STATUS LEDs



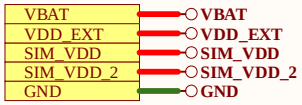


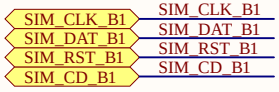


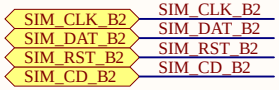










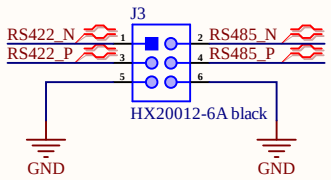


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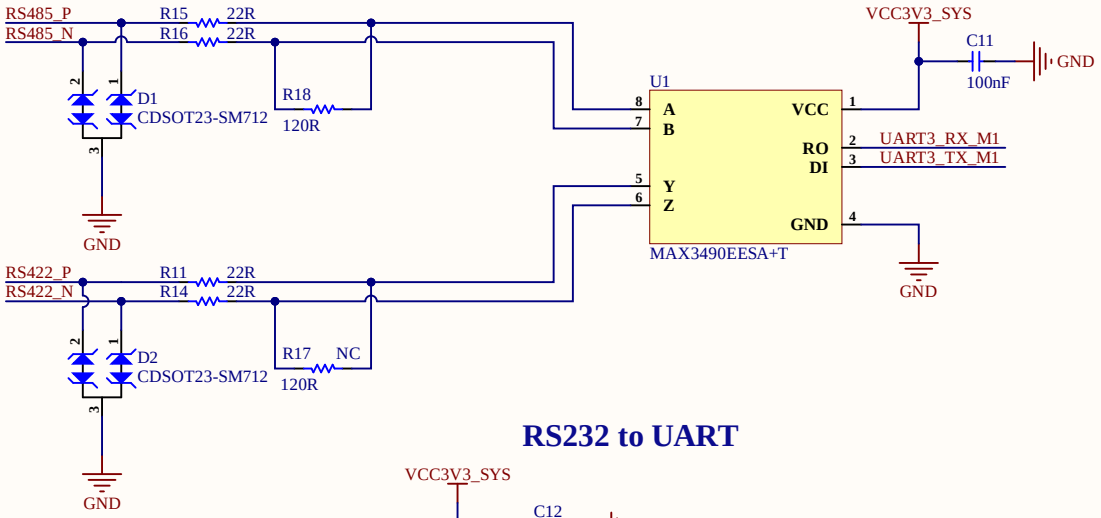
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RS232_RS485_INTERFACE

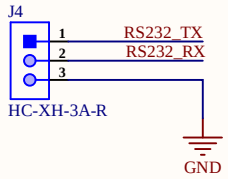
PORT for RS485



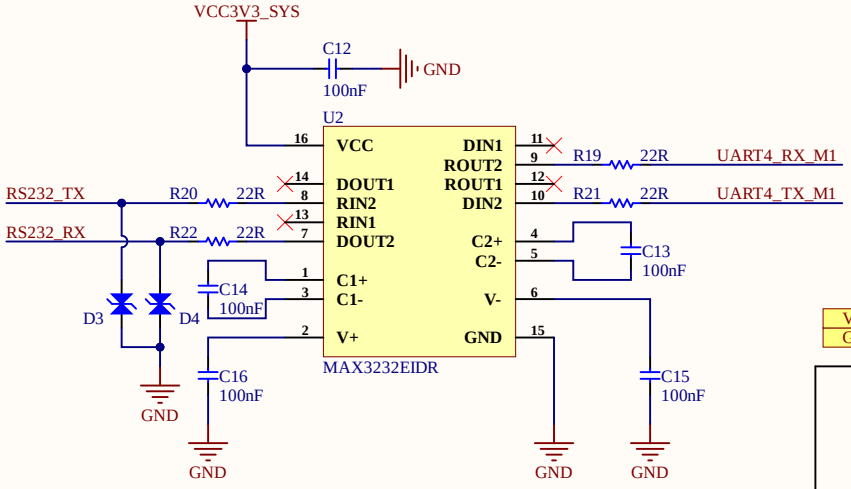
RS485 to UART



PORT for RS232



RS232 to UART



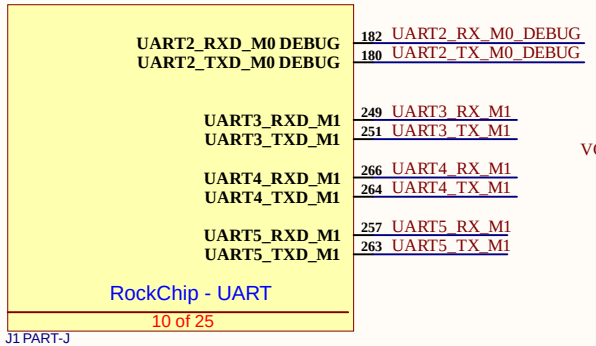


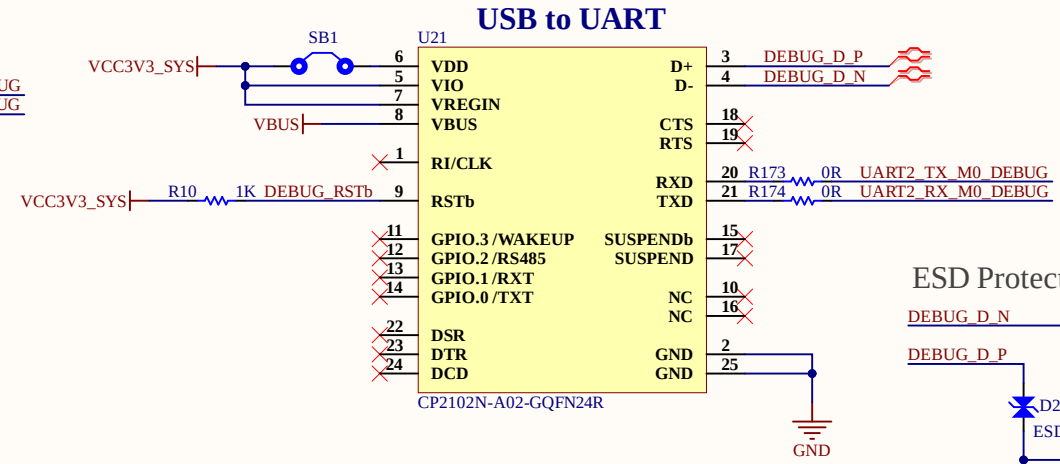


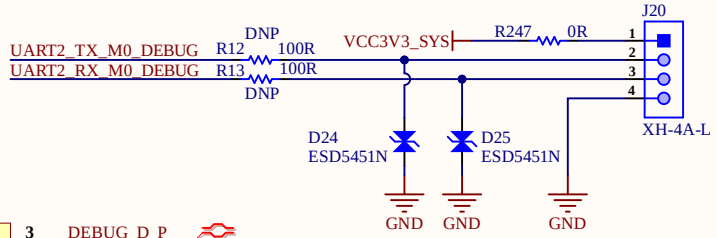
eSpark Consulting Group

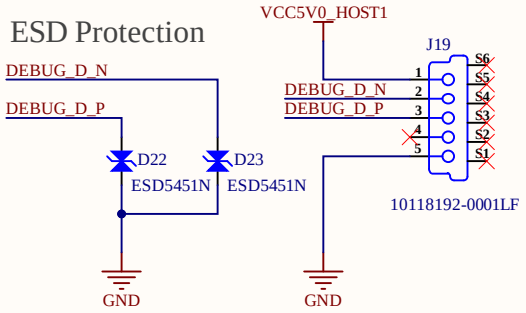
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Date: 12/06/2024	Page Num:	18
Drawn By: Huzaifa Ali	Page Total:	27

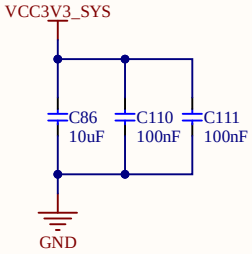
DEBUG_UART_INTERFACE

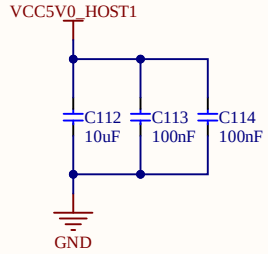


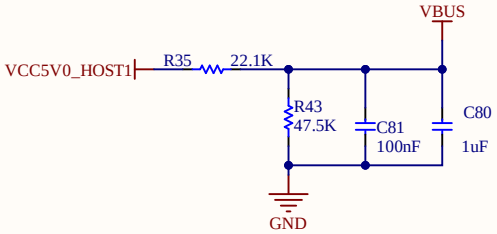


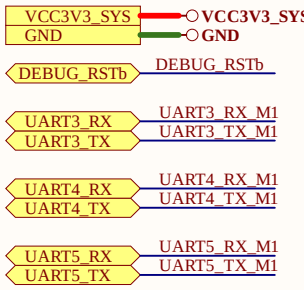










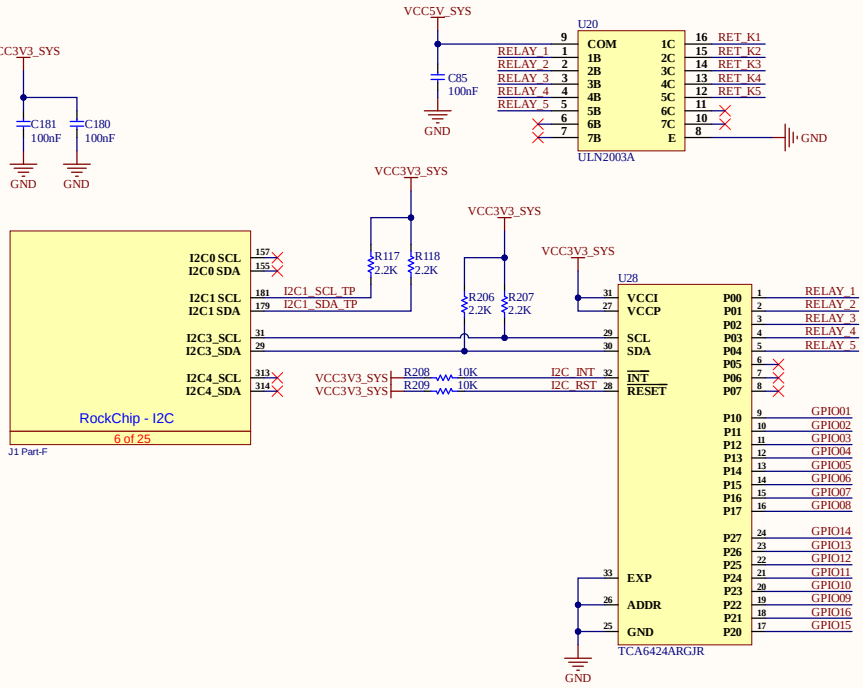


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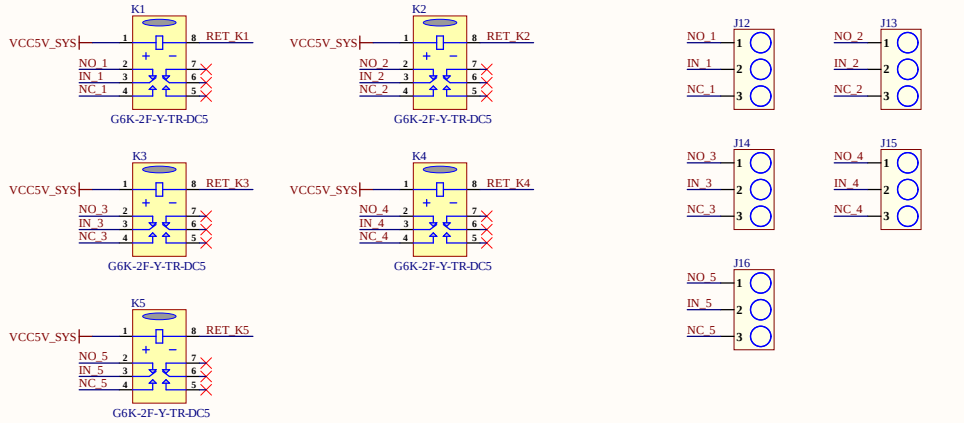
Title: DEBUG_UART_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 19
Drawn By: Huzaifa Ali	Page	Total: 27

DIGITAL IOs_RELAY_INTERFACE

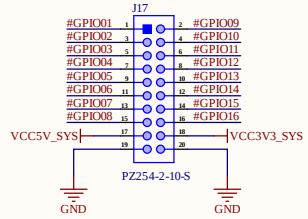
RELAY WITH DRIVER



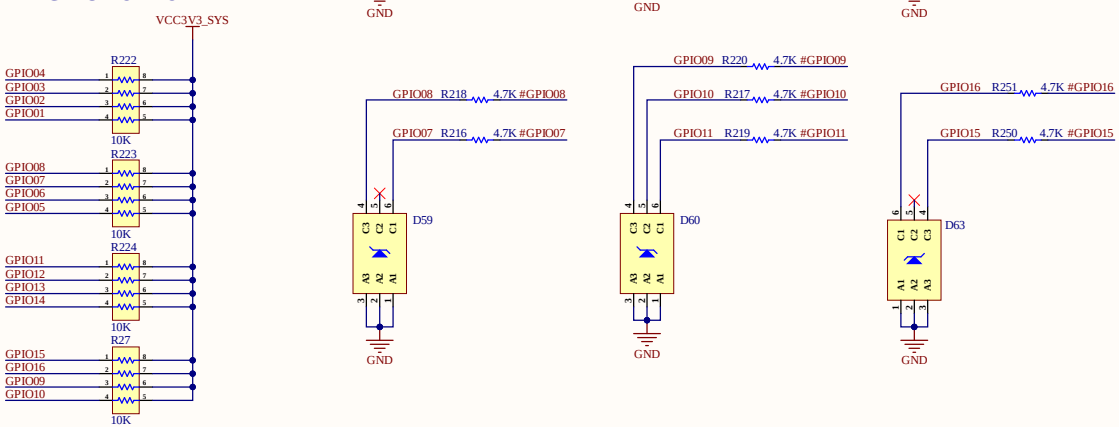
RELAY CONN

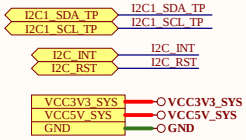


DIGITAL I/O



GPIO PULLUP

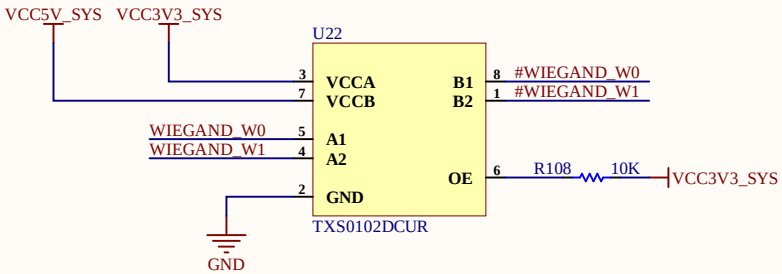


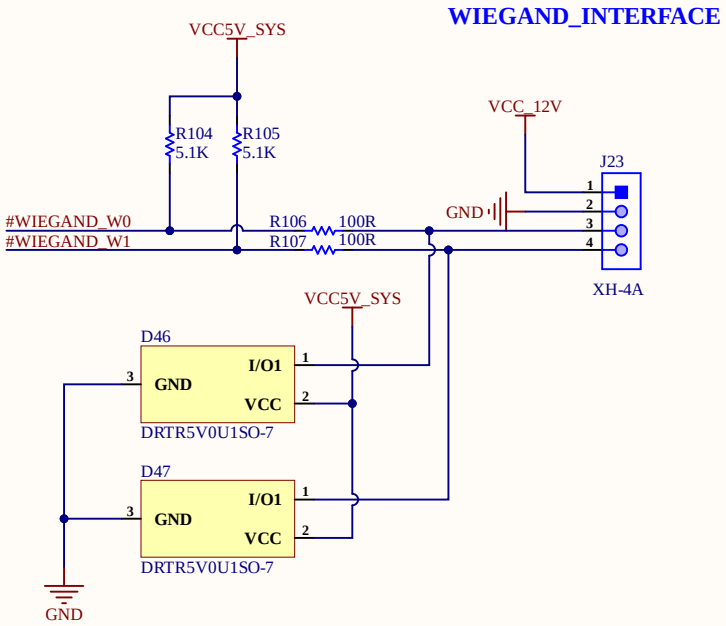


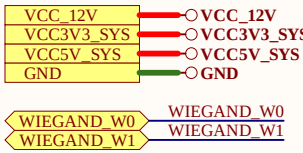
eSpark Consulting Group

Title: DIGITAL IO_RELAY_SHEET.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page	Num: 20
Drawn By: Huzaifa Ali	Page	Total: 27

WIEGAND_INTERFACE



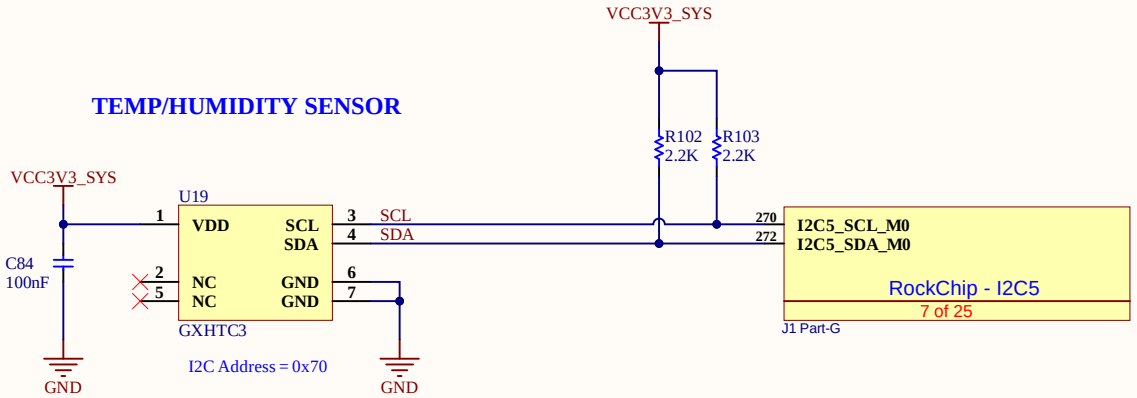




eSpark Consulting Group

Title: WIEGAND_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 21
Drawn By: Huzaifa Ali	Page	Total: 27

TEMP/HUMIDITY SENSOR_INTERFACE

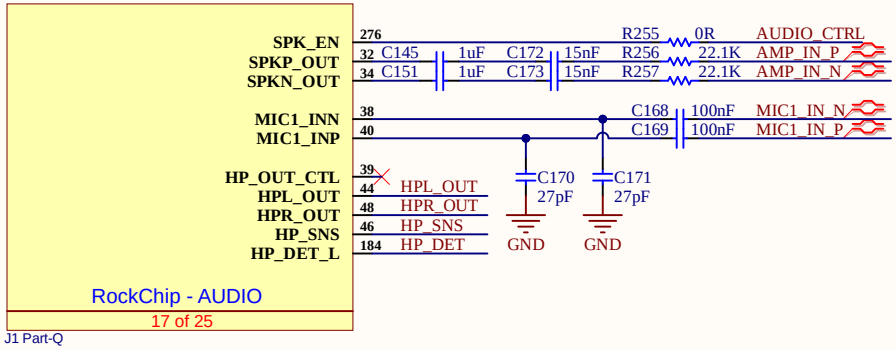


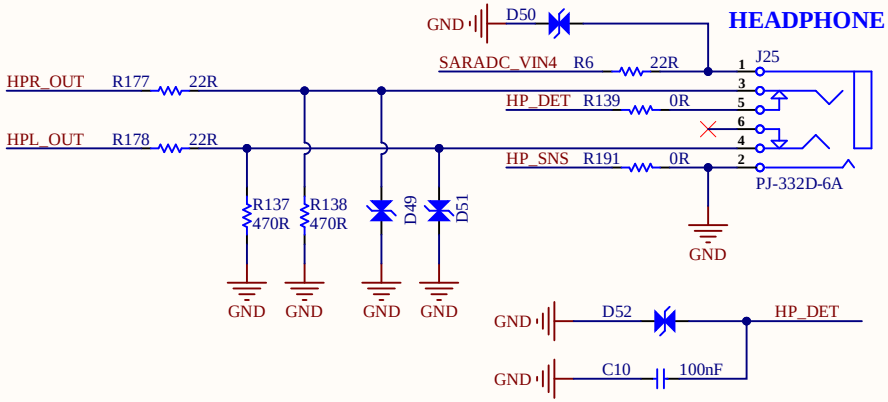


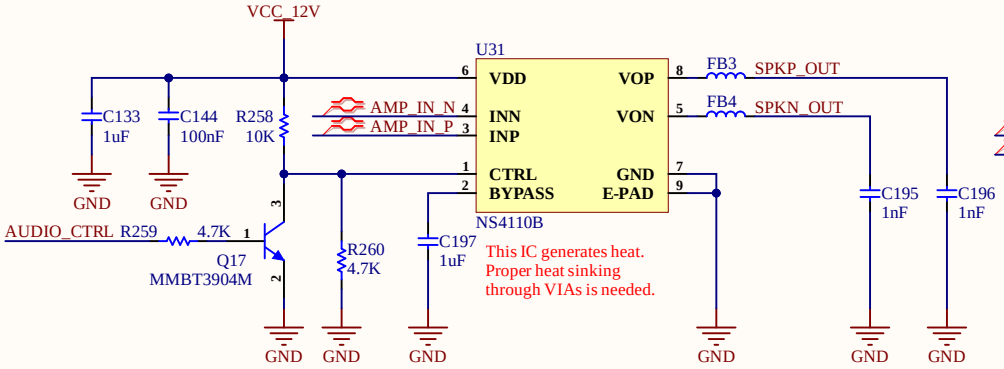
eSpark Consulting Group

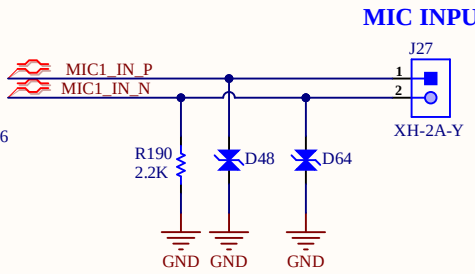
Title: SENSOR_SHEET.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 22
Drawn By: Huzaifa Ali	Page Total: 27

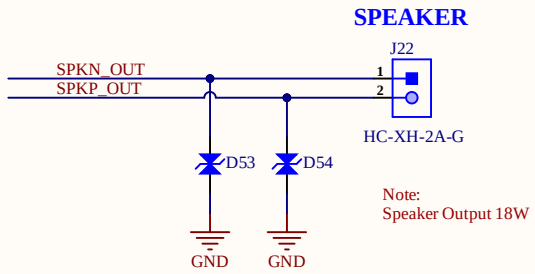
AUDIO_INTERFACE

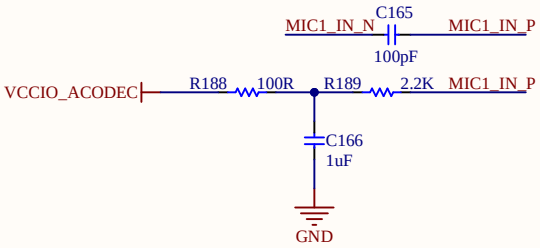


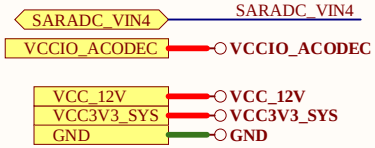










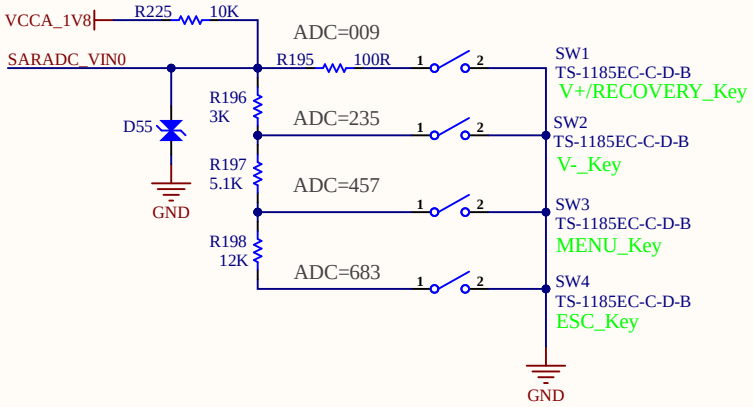


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Title: AUDIO_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 09/09/2024	Page Num:	23
Drawn By: Huzaifa Ali	Page Total:	27

PUSH BUTTON_INTERFACE

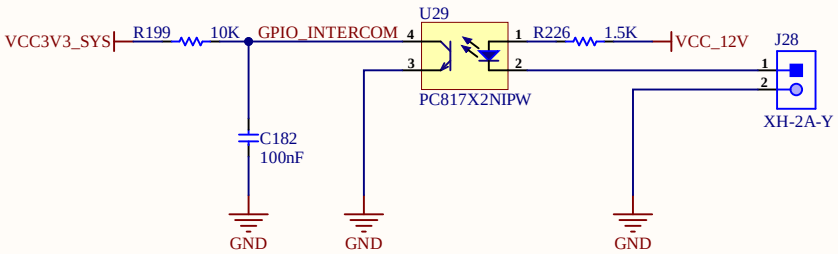
ACTION BUTTONs

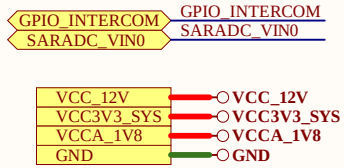


Note:

RECOVERY_Key function:
If SARADC_VIN0=0V
at after power on and reset,
then system will
enter into loader mode.

INTERCOM ACTION BUTTON

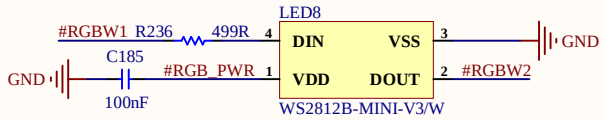


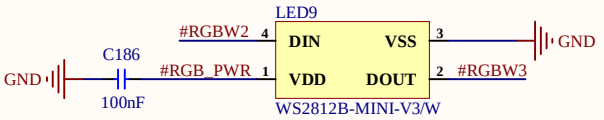


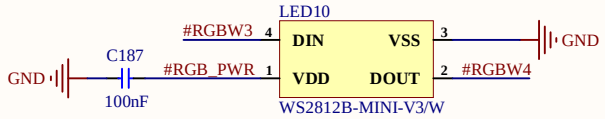
eSpark Consulting Group

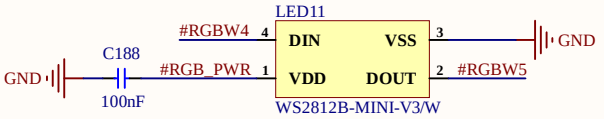
Title: PUSH BUTTONs_INTERFACE.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 24
Drawn By: Huzaifa Ali	Page Total: 27

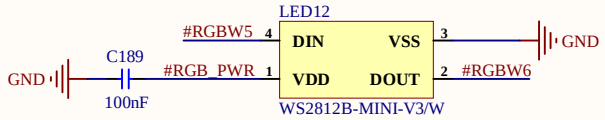
RGB_LEDs_INTERFACE

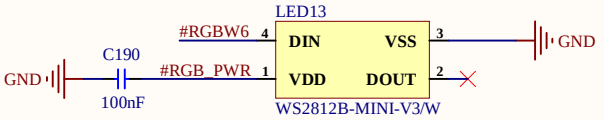


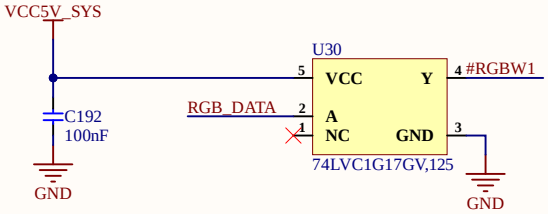


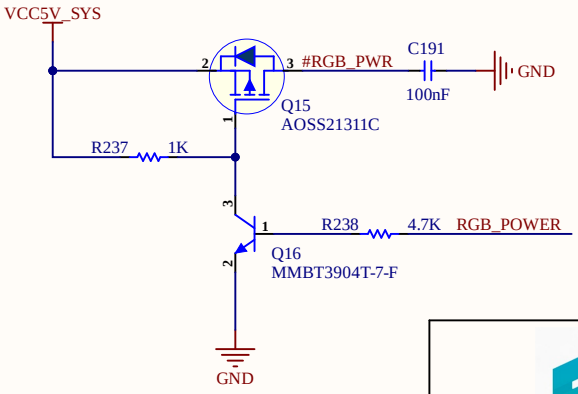












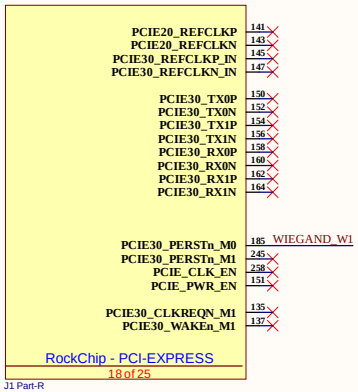




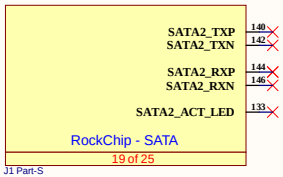
eSpark Consulting Group

Title: RGB_LEDs_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 25
Drawn By: Huzaifa Ali	Page	Total: 27

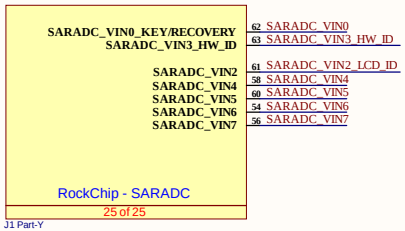
ROCKCHIP_GPIOs_NC PINs



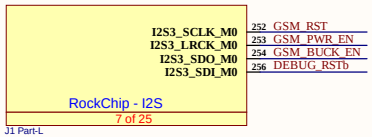
J1 Part-R



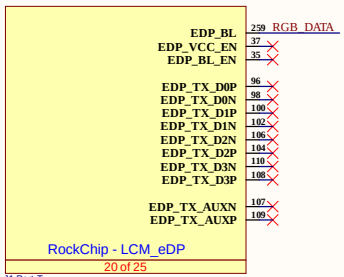
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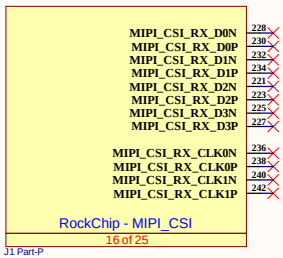
J1 Part-Y



J1 Part-L



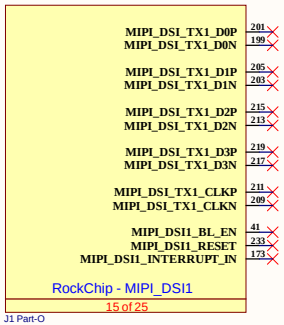
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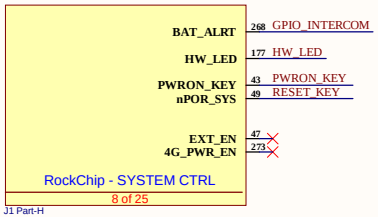
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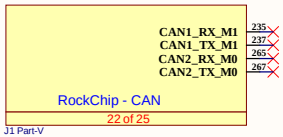
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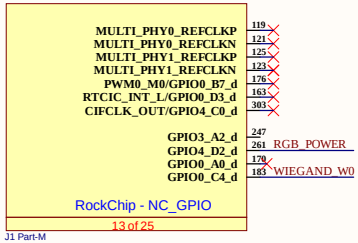
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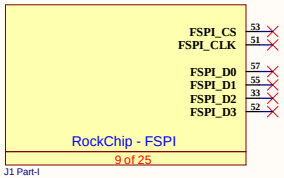
J1 Part-H



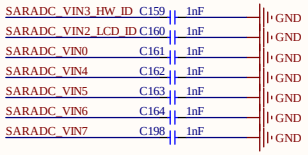
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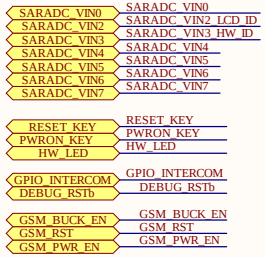
J1 Part-M



J1 Part-I







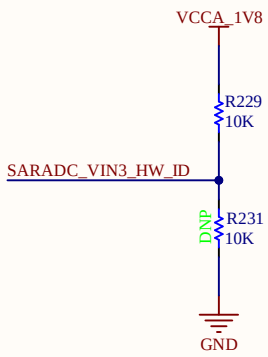
eSpark Consulting Group

Title: ROCKCHIP_GPIO_NC PINS.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/09/2024	Page	Num: 26
Drawn By: Huzaifa Ali	Page	Total: 27

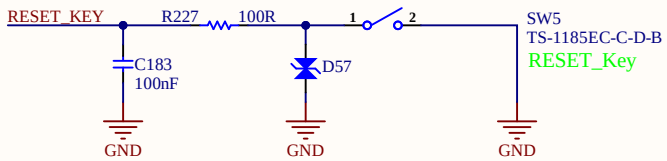
ROCKCHIP SYSTEM CONTROL

SARADC_VIN1	Up Resistance	Down Resistance
HW_ID0	10K	DNP
HW_ID1	10K	110K
HW_ID2	20K	100K
HW_ID3	33K	100K
HW_ID4	18K	36K
HW_ID5	36K	51K
HW_ID6	51K	51K
HW_ID7	51K	36K
HW_ID8	36K	18K
HW_ID9	100K	33K
HW_ID10	100K	20K
HW_ID11	110K	10K
HW_ID12	DNP	10K

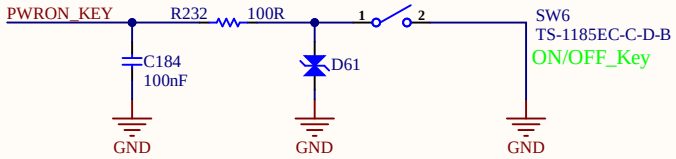
HW_ID0



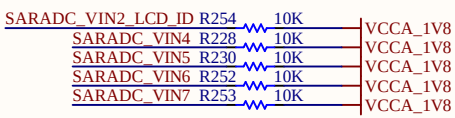
RESET_KEY



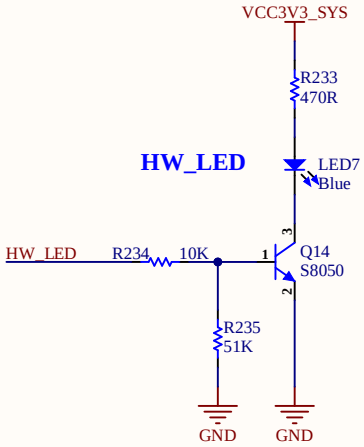
POWERON/OFF_KEY

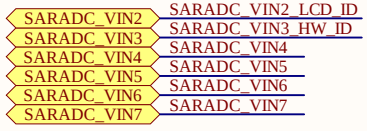


PULL-UP TO VCCA

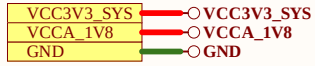


HW_LED









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Title: ROCKCHIP_SYSTEM_CONTROL.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 27
Drawn By: Huzaifa Ali	Page Total: 27

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.109mm	4.1	
2	L2_GND1	CF-004	0.030mm		
	Dielectric 4	PP-006	0.250mm	4.1	
3	L3_SIG1	CF-004	0.030mm		
	Dielectric 6	PP-006	0.109mm	4.1	
	Dielectric 8	PP-006	0.109mm	4.1	
4	L4_PWR	CF-004	0.030mm		
	Dielectric 1	FR-4	0.250mm	4.8	
5	L5_GND2	CF-004	0.030mm		
	Dielectric 9	PP-006	0.109mm	4.1	
	Dielectric 7	PP-006	0.109mm	4.1	
6	L6_SIG2	CF-004	0.030mm		
	Dielectric 5	PP-006	0.250mm	4.1	
7	L7_GND3	CF-004	0.030mm		
	Dielectric 3	PP-006	0.109mm	4.1	
8	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.674mm		

